

# RoboLink: Build and Drive Your Smartphone-Controlled Robot

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Analog Discovery 2

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## The Analog Discovery 2



## Introduction

Analog Discovery 2 (AD2) is a pocket size instrument from Digilent that can fairly replace an entire electronics lab, within low limits in frequency, voltage, and power. It can work as oscilloscope, waveform generator, voltmeter, DC power supply, spectrum analyzer, network analyzer, logic analyzer, and much more. In the following experiment, we will use the AD2 to read the I2C signals between the sensor and the Arduino Due board to check whether the sensor is working properly or not.

## How it works

In typical configuration, AD2 lies between the circuit under inspection and a computer as shown in Fig. 5.1. We connect wires and/or coaxial cables (with BNC connectors) to carry analog/digital signals from and to the circuit under testing. Meanwhile, the AD2 is connected to a computer by a USB cable to carry commands and configuration parameters to AD2 and reply with measurement values from the AD2 to the computer. Digilent provides a software called “WaveForms” that works as a GUI for the AD2 among a group of similar supported instruments. Through WaveForms the user can do the required configurations to any module on the AD2, e.g., select trigger source for oscilloscope, waveform type for waveform generator, ...etc. The software also provides graphical display to show the measured quantities in real time (Oscilloscope, spectrum analyzer, ...etc.). User scripts can be also implemented (using JavaScript) on WaveForms to customize the

AD2 tasks, which adds a great flexibility to its performance to match the user requirements. Digilent also provides WaveForms SDK with extensive API that allows controlling the AD2 and creating custom PC applications. The SDK allows users to use different programming languages like C/C++, C#, Python, and Visual Basic and provides an extension to Matlab runtime environment.

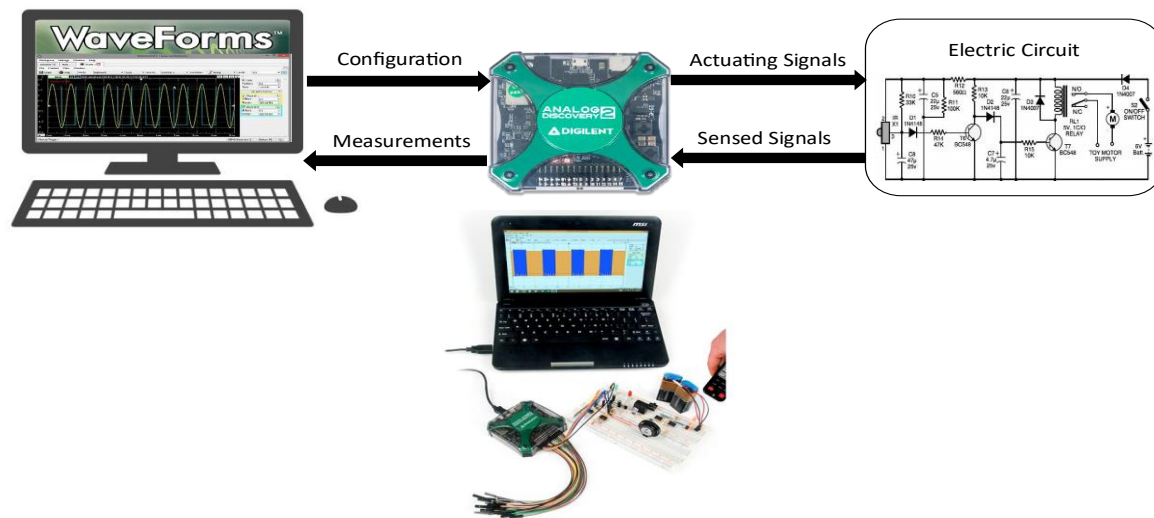


Fig. 5.1 Working Concept of Analog Discovery 2

## Specifications

For the circuit side, the AD2 provides the following interfaces as shown in Fig. 5.2:

- 1- **Analog Input Interface:** Two independent analog input channels are provided. These two channels are shared between multiple modules in the AD2 like the scope and voltmeter.
- 2- **Analog Output Interface:** Two independent analog outputs channels are provided. The two output channels are used mainly by the waveform generator, but they are also used by other modules like the network analyzer.
- 3- **Digital IO Interface:** 16 digital IO interfaces are provided. They are used mainly for the logic analyzer module, and they are also shared with more advanced functions like serial protocols analyzers.
- 4- **Power Supply Interface:** Two power supply interfaces are provided, a positive power supply with range 0.5V to 5V and a negative power supply with range -5V to -0.5V.
- 5- **Trigger Interface:** Two external trigger inputs for the scope functionality.

In the following, a summary of the specs of the commonly used modules in the AD2 is presented. Please refer to the reference manual [Analog Discovery 2 Reference Manual - Digilent Reference](#) for more detailed information about all modules.

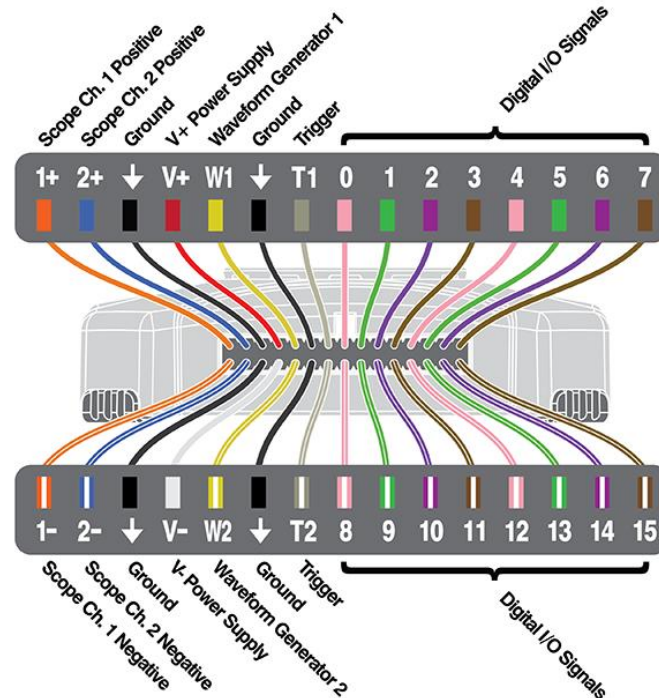


Fig. 5.2 Pins description of AD2

### As a 100 MS/s Oscilloscope Scope

The AD2 is used with WaveForms' Oscilloscope instrument to capture analog input data via the analog input ("Scope") channels, using either BNC cables or jumper wires. When this instrument is used, the AD2's analog input channels act as a two-channel, 14-bit, 100 MS/s oscilloscope. A screenshot from typical scope view is presented in Fig. 5.3.

When used with jumper wires, the oscilloscope channels' + and - pins are differentially paired. The - pins can be attached to a non-ground circuit net, but the circuit under test must still share a common ground with the AD2. However, when using BNC cables, the oscilloscope channels are single-ended and the circuit under test must still share a common ground with the AD2. With BNC cables, the analog input channels have a substantially higher bandwidth than with jumper wires.

In the following experiment, we will only use the jumper wires provided with the kit. We provide an introduction for using the AD as an Oscilloscope as a quick debug tool which can replace the

benchtop Oscilloscope while you develop the term project. This is only possible as the clock frequency of the Arduino Due board is relatively small (84 MHz).

Since the AD2's analog input channels are shared, the Oscilloscope instrument cannot be used at the same time as the Voltmeter, Data Logger, Spectrum Analyzer, Network Analyzer, or Impedance Analyzer instruments.

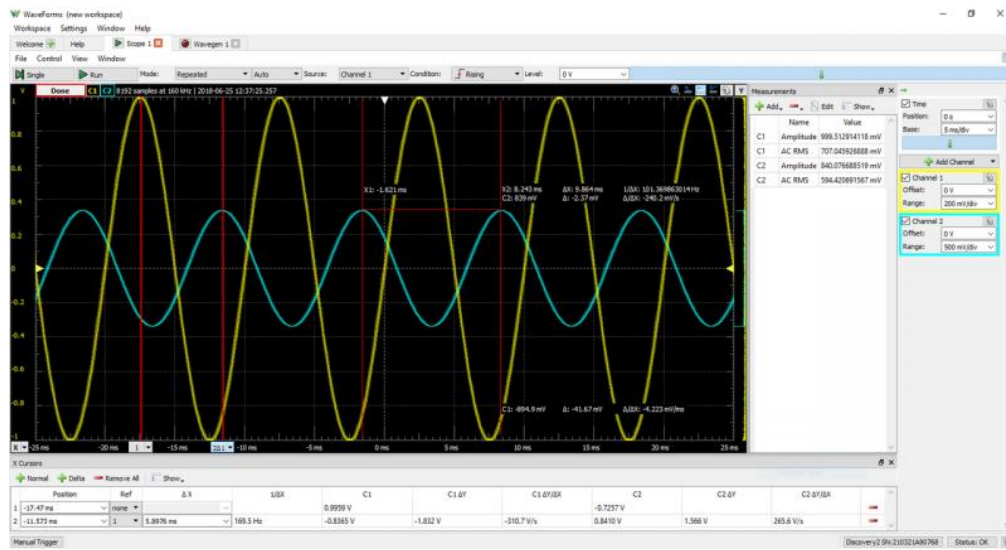


Fig. 5.3 Screenshot from Scope module in WaveForms

### Waveform generator

WaveForms' Wavegen instrument outputs analog voltage waves via either BNC or Jumper wires. Wavegen converts 14-bit digital samples to analog at a rate of up to 100 MS/s on each of two channels. When Wavegen is used, the AD2's analog output channels act as an Arbitrary Waveform Generator. The instrument supports everything from simple waveforms like Sine and Triangle waves, up to more complicated functions like AM and FM modulation. Custom sets of samples can be defined by the user in applications like Excel and imported to WaveForms. Each waveform generator channel is considered a single ended pin; however, a connected circuit must share a ground with the Analog Discovery 2. A screenshot from the module's window is provided in Fig. 5.4

Since the AD2's analog output channels are shared, the Waveform Generator instrument cannot be used at the same time as the Network Analyzer, or Impedance Analyzer instruments.

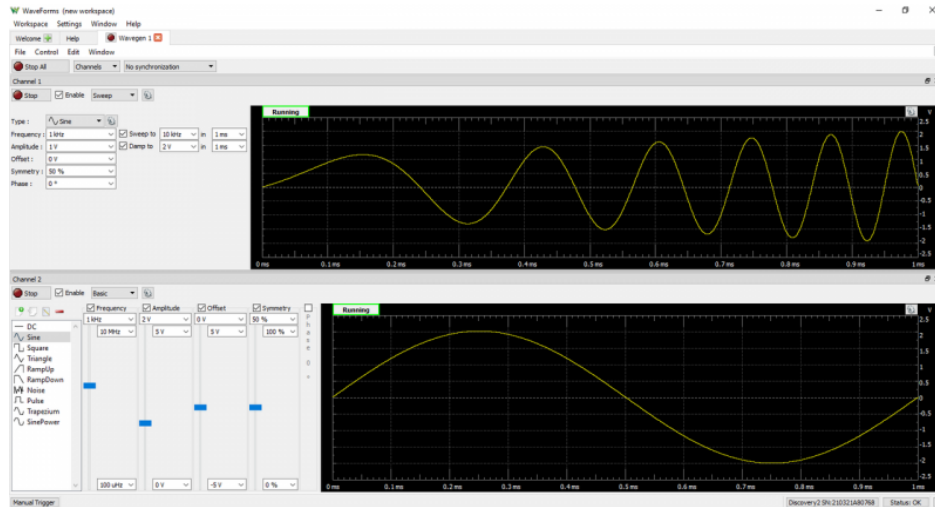


Fig. 5.4 Screenshot from Wavegen interface in WaveForms.

### As a voltage supply

The AD2 can be used to generate a DC volt that can be used as a power source to other circuits. However, it should be noted that its current supply is limited by its power source.

### Required Equipment

- Analog Discover 2
- Jumper wires.

### Procedure

- 1- Connect the Analog Discovery 2
- 2- From the Welcome screen, select the Supplies tool
- 3- Make sure the Master Enable is OFF.  Master Enable is Off .
- 4- Activate the V+ channel to green check  Positive Supply (V+) Rdy .
- 5- Deactivate the V- channel  Negative Supply (V-) Off .
- 6- Set the output voltage of the V+ channel to 5V (or as required).
- 7- Use the V+ pin and any ↓ ground pin from the AD2 as your power terminals.
- 8- Activate the Master Enable.

# Experiments

In the following experiments, we will try to investigate the basic functionality of the AD2 and WaveForms application to get familiar with the instrument.

### Experiment 1: Test the AD2 by Loopback

In this experiment, the AD2 will produce a pre-configured waveform on the analog outputs of the AD2 and measure the analog signals back into the scope interface. We target three goals: 1- test the AD2 as a standalone unit, 2- test the AD2 as a function generator, and 3- test the Oscilloscope functionality of the AD2.

#### Require Equipment

- Analog Discover 2
- Jumper wires.

#### Procedure

- 1- Download, install and run the Digilent WaveForms from:

<https://digilent.com/shop/software/digilent-waveforms/>

Remember to install the USB drivers.

- 2- Make connections:

- a. +1 pin to W1 pin.
- b. +2 pin to W2 pin.
- c. -1 and -2 to the ↓ Ground pin.

- 3- From the Welcome Screen, select the wavegen module of the Digilent WaveForms software.

Activate the two channels, by selecting them from the Channels menu.

Configure the wavegen channels to generate the following.

- a. Channel 1: A sine wave with frequency of 1 KHz and 1 V amplitude.
- b. Channel 2: A square wave with period of 500 usec (frequency of 2kHz), the minimum value is 0V, the maximum value is 2V, and the duty cycle (symmetry) is 25%.

- 4- From the Welcome screen, select the Scope. Configure the scope module as follows

- a. Source of trigger (at the top) to be Channel 1

- b. Time scale to display 5 periods of signal 1 on horizontal axis, and 500 mv per division for vertical axes of both signals.

**Bonus:** From the Welcome screen, use the Specturm tool to display the FFT of both signals, does it match your expectations?

### Experiment 2: Using AD2 as a voltmeter

In this experiment, we will use the AD2 as a voltmeter to measure the voltages in our circuits.

#### Required Equipment

- Analog Discover 2
- Jumper wires.

#### Procedure

- 1- In WaveForms, go to the Voltmeter tab to display the readings.
- 2- Click “Run”
- 3- Use +1, -1 terminals from the AD2 to measure voltage.
  - a. You can measure the voltage of the power supply you managed to setup eraluer.